RANJAN KUMAR SAHOO

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CAREER OBJECTIVE

To apply my knowledge of software development, backend engineering, Artificial Intelligence, and Computer Vision to develop innovative solutions, contribute to organizational success, and continuously enhance my technical expertise.

EDUCATION

Master In Technology (Computer Science & Engineering) | 9.1 CGPA — June 2025 – May 2027

Alliance University, Bangalore, Karnataka

Master In Computer Application | 8.4 CGPA — June 2019 – Sept 2022

Biju Patnaik University of Technology, Raurkela, Odisha

Bachelor in Science (Information Technology and Management) | 7.1 CGPA — June 2016 – June 2019

Utkal University, Bhubaneswar, Odisha

TECHNICAL SKILLS

Systems & Backend: Linux, windows, Java 8, Spring Boot, Microservices, Docker, AWS, REST APIs, MySQL, MSSQL, Git

ML & AI: Python, TensorFlow, Scikit-learn, OpenCV, DeepFace

Generative AI & RAG: RAG pipelines, vector databases (Qdrant), semantic search, embedding models

WORK EXPERIENCE

Junior Software Developer — Dec 2022 –Dec 2024

Internet Research Bureau, Noida, U.P.

- Project Name : OSBT (Online Survey Bureau Tool)
- Domain - https://onlinesurveybureau.com/
- Technology - Java 8, Spring boot, JUnit, Event-Driven Architecture, MS_SQL, GIT, STS, PUTTY, Agile methodology
- Coding, Unit Testing, Implementation and Documentation and deploy into the web server
- Objective: Create a robust generic product to filter the user and take the perfect survey into the perfect person as per the craiteria and my responsibilities are to develop different modules from scratch to filter and store in-complete transactions in the CSRQ pattern.

- Project Name - Sales pasmt (panel and survey management tool in sales)
- Domain - http://sales.pasmt.com/
- Technology - Java 8, Spring MVC, JUnit, MS_SQL, GIT, STS, PUTTY, Agile methodology
- Coding, Unit Testing, Implementation and Documentation and deploy into the web server
- Objective - This is a tool where different different user like sales manager, bid manager create biding for the project and create there proposal letter for the client and manage the project sales, finance

Web Development Internship — Oct 2021 – April 2022

Pepcodeing Education PVT. LTD, Noida, U,P.

- Learning Full stack web development and learning technology like JAVA, Springboot and Node JS to improve my skill and knowledge.

PUBLICATION

Sahoo, R.K. & Shekhar, R. *"Batch Edge Updates for Fully Dynamic All-Pairs Shortest Paths in the MPC Model."* Under Review — International Conference on Advances in Computing Systems, Communication & Networking.

RESEARCH & ACADEMIC PROJECTS

Zero-Shot Affordance Detection Under Occlusion via Occlusion-Aware Training on Raw Point Clouds (Ongoing research)

- Developed a geometry-augmented DGCNN that predicts per-point affordances (grasp, contain, wrap-grasp, pour) on 3D point clouds using 7D input features (XYZ + estimated surface normals + curvature), achieving 20.4% mIoU on seen categories and demonstrating non-trivial zero-shot transfer (9.4% mIoU) to unseen object categories.
- Designed a realistic viewpoint-based occlusion augmentation strategy (50% training probability) that maintains 92% of clean-view performance under simulated partial observations, validated across a 4-condition evaluation protocol (seen/unseen × clean/occluded) on the 3D AffordanceNet benchmark (23 categories, 16K samples).

Placement Preparation RAG System

Backend: FastAPI (Python) | Vector DB: Qdrant (Docker) | Frontend: React

- Built an end-to-end Retrieval-Augmented Generation system allowing users to upload PDFs, text, and Markdown into named collections and receive cited answers.
- Implemented semantic search over vector embeddings (SentenceTransformers + Qdrant) with multi collection querying and citation attribution; added a quiz mode for active recall over ingested materials.

AI-Powered Face Detection Attendance System

- Developed a real-time attendance system (Python, TensorFlow, OpenCV, DeepFace); analysed robustness under noisy real-world conditions-varying lighting and occlusion connecting to dependable ML themes.

CERTIFICATES

Tools of the Trade: Linux and SQL
- Google certificate

Generative AI: Prompt Engineering
- IBM certificate

LANGUAGES

English | Hindi | Odia